Grassmann Schemes: Permutation Molien Formula Proof and Exact Verification

Executable companion note

17 July 2026

Abstract

For $GL_N(q)$ acting on the k-subspaces of $\mathbb{F}_q^N$, we prove the permutation Molien formula and verify its invariant-subspace input by direct prime-field matrix construction. Three representative cases test points, hyperplanes, and a nontrivial scalar kernel.

1 General formula

Let $X = \mathrm{Gr}_q(N, k)$ and $V = \mathbb{C}[X]$. If $c_\ell(g)$ is the number of ℓ-cycles of g on X, the cycle-block determinant identity gives

$$\mathcal{S}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i,\ell} (1 - t_i^\ell)^{-c_\ell(g)}. \tag{1}$$

The monomial basis of $\mathrm{Sym}^d(V)$ is indexed by d-multisets of subspaces, so Burnside's lemma also proves

$$[m_\tau]\mathcal{S}_{G,X} = \#\left(G \backslash \prod_j \mathrm{Sym}^{\tau_j}(X) \right). \tag{2}$$

For a linear matrix g, define

$$F_r^{\mathrm{Gr}}(g) = \#\{U \le \mathbb{F}_q^N : \dim U = k,\ g^r U = U\}. \tag{3}$$

Every ℓ-cycle contributes its ℓ vertices to F_r exactly when $\ell \mid r$. Möbius inversion therefore yields

$$c_\ell^{\mathrm{Gr}}(g) = \frac{1}{\ell} \sum_{d \mid \ell} \mu(\ell/d) F_d^{\mathrm{Gr}}(g). \tag{4}$$

Together, (1), (3), and (4) prove the claimed exact formula. Interpreting the ambient space as an $\mathbb{F}_q[x]$-module with x acting as g^r identifies (3) with the stated invariant-submodule problem.

Pair orbits are classified by $\dim(U \cap W)$, and so, with $s = \min(k, N - k)$,

$$[m_{(1)}] = 1, \qquad [m_{(2)}] = [m_{(1,1)}] = s + 1.$$

2 Verification strategy

All matrices in $GL_N(q)$ are enumerated over the prime field. Every k-subspace is generated and stored in unique reduced-row-echelon form. Matrix multiplication then gives both the induced vertex permutation and the independent invariance test in (3). Scalar matrices are deduplicated before averaging, so the average is over the actual permutation image; this is equivalent to averaging over $GL_N(q)$ because the scalar kernel has constant size.

The target coefficient is computed from actual permutation matrices using Newton's trace recurrence for $h_d(P) = \mathrm{tr}(\mathrm{Sym}^d P)$. The same coefficient is independently extracted from (1) and counted as the explicit orbit number in (2). Finally, invariant-subspace counts for powers of every matrix representative must reconstruct every cycle through (4).

3 Exact results

The coefficient vector is ordered by $\tau = (1), (2), (1,1), (3), (2,1), (1,1,1)$.

Action	$\lvert GL\rvert$	$\lvert G$ image$\rvert$	$\lvert X\rvert$	common vector
$\mathrm{Gr}_2(3,1)$	168	168	7	$(1,2,2,4,5,6)$
$\mathrm{Gr}_2(3,2)$	168	168	7	$(1,2,2,4,5,6)$
$\mathrm{Gr}_3(2,1)$	48	24	4	$(1,2,2,3,4,5)$

All 18 exact triple comparisons and all invariant-subspace reconstructions pass. The final row explicitly checks the scalar kernel. In the first case, the direct determinant and cycle averages at $(0.037, 0.061)$ both equal 1.1152638116688691 (reported error 0).

4 Reproduction

Run

```
marimo run for_this_guy/association_scheme_permutation_representations/
  grassmann/notebook.py
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from the repository root. The implementation is intentionally restricted to prime fields for transparency; the theorem itself is not so restricted.